MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Customer: **Mansourah Massarah Gold Project**
MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: **GENERAL ARRANGEMENT DRAWING**
FUEL STORAGE FACILITY - MISCELLANEOUS
PIPE SUPPORT & EQUIPMENT FOUNDATION

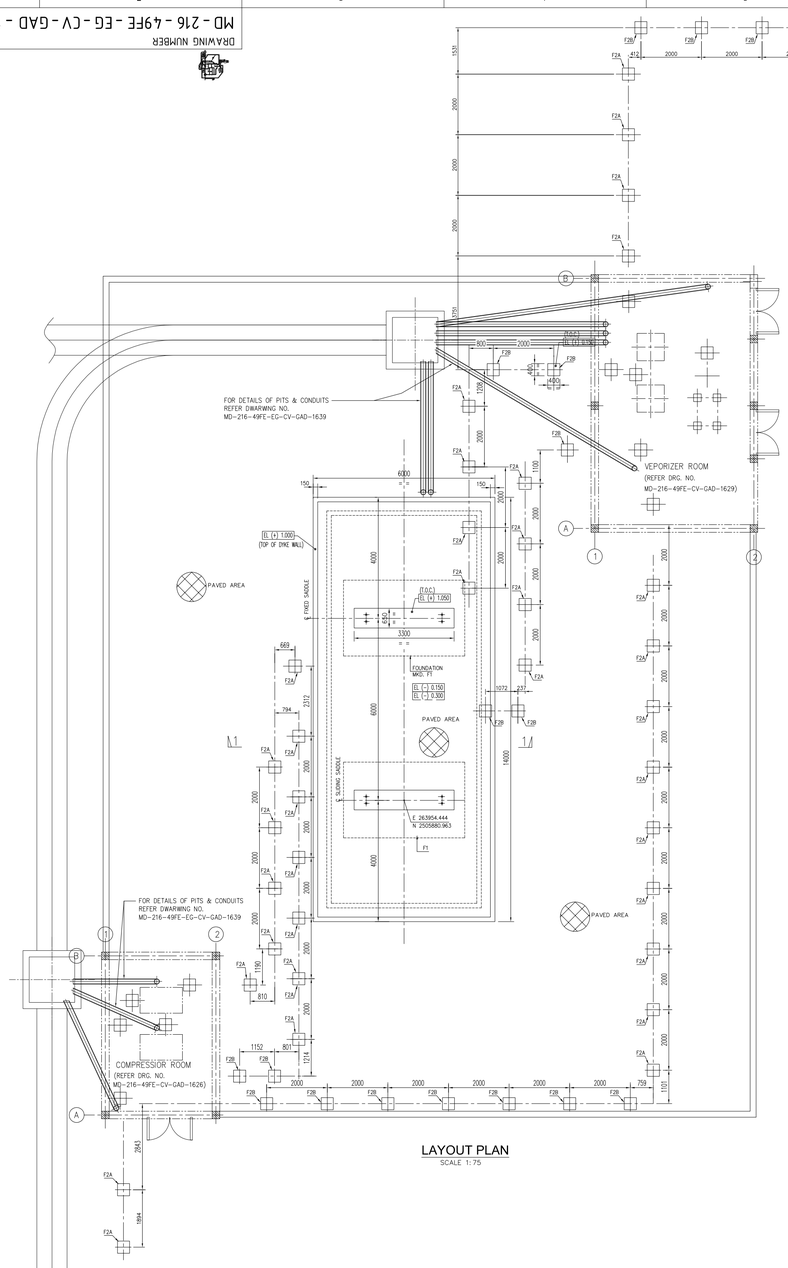
Document Number: **MD-216-49FE-EG-CV-GAD-1631**
Revision: **D01**
Revision Date: **22.07.2023**
Status: **AS BUILT**

Outotec Document Number:	900707	GAU01	AG50	MTL01	49037	OU100703980	4
L&T Document Number:	O19045	080	C	G	24313	004	4

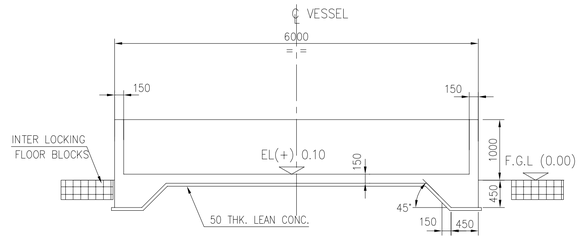
REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	19.10.2020	ISSUED FOR APPROVAL	TEC	SJM	AG	
C01	27.11.2020	ISSUED FOR CONSTRUCTION	TEC	SJM	AG	
C02	14.04.2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. C02	TEC	SJM	AG	
C03	28.09.2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. C03	TEC	SJM	AG	
D01	22.07.2023	AS BUILT	TEC	SJM	AG	

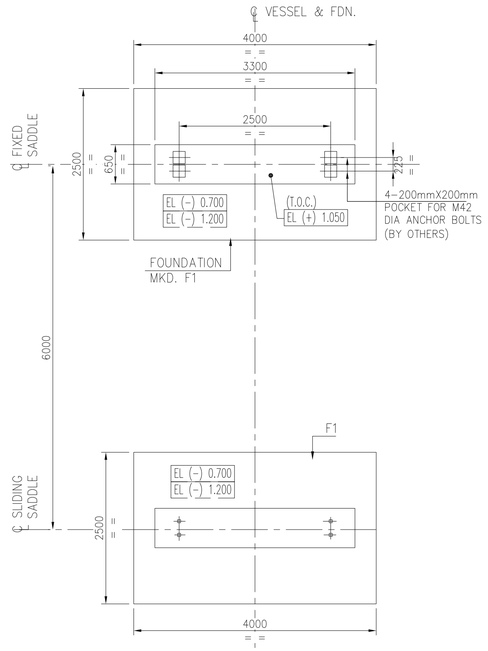
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MD-216-49FE-EG-CV-GAD-1631
REV D01



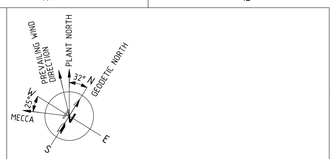
LAYOUT PLAN
SCALE 1:75

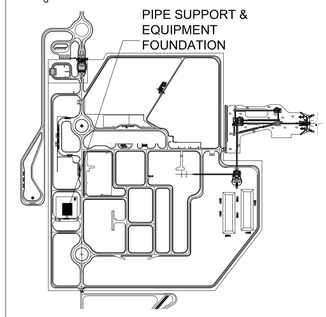


SECTION 1-1
TYP. CROSS SECTION OF DYKE WALL
SCALE 1:50



FOUNDATION LAYOUT PLAN
SCALE 1:50





KEY PLAN

LEGENDS

F-FOUNDATION
CR-COLUMN NECK
C-COLUMN
FFL-FINISH FLOOR LVL
FGL-FINISH GRADE LVL
TOS-TOP OF SCREED
WWF-WELDED WIRE FABRIC
HCS-HOLLOW CORE SLAB
GB-GRADE BEAM
IJ-ISOLATION JOINT
CJ-CONTROL JOINT
GA-GALVANIZED
S.O.G-SLAB ON GRADE
ARCH-ARCHITECTURAL
HOR.-HORIZONTAL
REINF.-REINFORCEMENT
LVL.-LEVEL
MKD.-MARKED
Ø- DIAMETER
TYP.-TYPICAL
THK.-THICKNESS
T/O BEAM-TOP OF BEAM
C/C-CENTRE TO CENTRE
℄-CENTRE LINE
EL.-ELEVATION
SPECS.-SPECIFICATION
DOC.-DOCUMENT
T.O.C.-TOP OF CONCRETE
DRG.-DRAWING

NOTES

1. ALL DIMENSIONS ARE GIVEN IN MILLIMETRES AND ELEVATIONS ARE IN METRES UNLESS OTHERWISE NOTED.
2. MINIMUM REINFORCED CONCRETE GRADE TO BE USED FOR SUPERSTRUCTURE & SUBSTRUCTURES IS M35(5000 PSI).
3. MINIMUM LEAN CONCRETE GRADE TO BE USED FOR THE SUB-STRUCTURES IS M15(2175 PSI).
4. ALL REINFORCING STEEL SHALL BE DEFORMED BILLET STEEL BARS CONFORMING TO ASTM-A615, WITH MINIMUM YIELD STRENGTH OF FY420 MPA .
5. THE NET SOIL BEARING CAPACITY CONSIDERED IN FOUNDATION DESIGN IS 350 KPA(REFER DOC 216-3000-EG-CV-CAL-1008-A02).
6. WATER PROOFING INSULATION SHALL BE AS PER PROJECT SPECS AND ITS MATERIAL TO BE APPROVED BY ENGINEER AT SITE.
7. THE MINIMUM CONCRETE COVER AS PER SBC-304-2018(TABLE 20.6.1.3.1) MEASURED FROM REBAR SHALL BE
   a) CONCRETE CAST AGAINST & PERMANENTLY EXPOSED TO EARTH (INCLUDING FOUNDATIONS OVER SUB-SLAB)
      FORMED CONCRETE EXPOSED TO EARTH OR WEATHER.....50MM
   b) CONCRETE NOT EXPOSED TO WEATHER NOR IN CONTACT WITH GROUND
      BEAMS, COLUMNS.....40MM
      SLABS, WALLS, JOISTS.....20MM
   c) CONCRETE SLAB OVER 50MM SUB-SLAB.....20MM
8. ALL DRAWINGS MUST BE READ IN CONJUNCTION WITH RELEVANT ARCHITECTURAL AND ENGINEERING DRAWINGS RELATED TO THE SAME SHEET AND ANY ITEM ON IT.
9. F.F.L +0.00 MENTIONED IS EQUIVALENT TO FINISH FLOOR LVL +913.50M AS PER THE CIVIL DRAWING CE-004A-OVERALL LAYOUT PLAN.

SHEET 1 OF 2

DOCUMENT NUMBER	REFERENCE DOCUMENT	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	20/7/2023	AS BUILT	TEC	AVG	TEC	SJM	
		C03	28/09/2021	ISSUED FOR CONSTRUCTION & REVISED AS MARKED △C03	TEC	AVG	TEC	SJM	
		C02	14/04/2021	ISSUED FOR CONSTRUCTION & REVISED AS MARKED △C02					
MD-216-6000-EG-CV-STD-1002	HOLLOW CORE SLAB STANDARD	C01	27/11/2020	ISSUED FOR CONSTRUCTION	TEC	SJM	TEC	AG	
MD-216-6000-EG-CV-STD-1001	REINFORCEMENT STANDARD	A01	19/10/2020	ISSUED FOR APPROVAL	TEC	SJM	TEC	AG	

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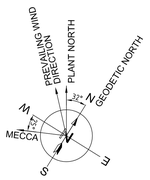
LARSEN & TOUBRO DRAWING NUMBER: 019045_080_C_G_24313_004 — REVISION: 2
OUTOTEC DRAWING NUMBER: 900707__GAU01__AG50__MTL01__49037 — DOCUMENT ID: OU100703980 — REVISION: 2

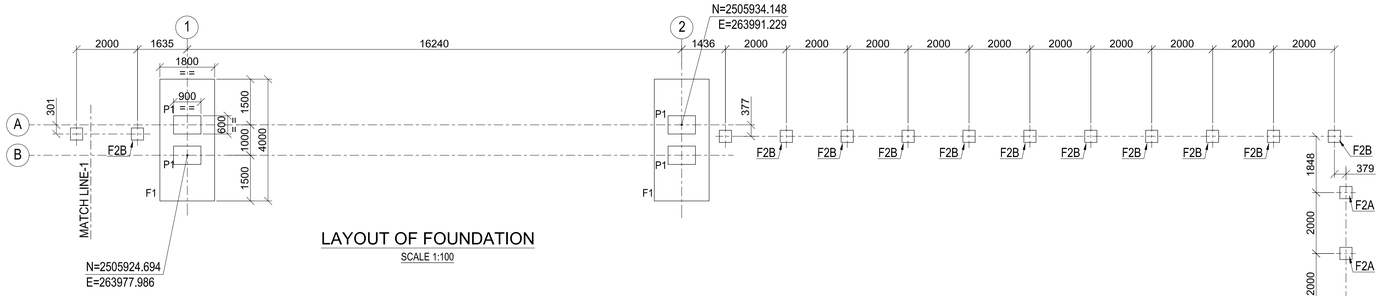
DRAWN	TEC	19/11/2020
CHECKED	SJM	20/11/2020
ENGINEER	TEC	19/11/2020
APPROVED	AG	20/11/2020
EMPLOYER		
SCALE	1:50,10	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
FUEL STORAGE FACILITY-MISCELLANEOUS
PIPE SUPPORT & EQUIPMENT FOUNDATION

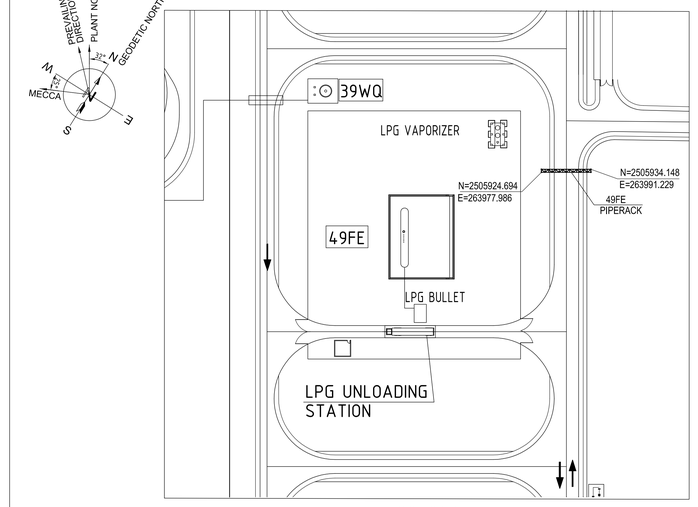
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DRAWING NUMBER
MD - 216 - 49FE - EG - CV - GAD - 1631
REV
D01

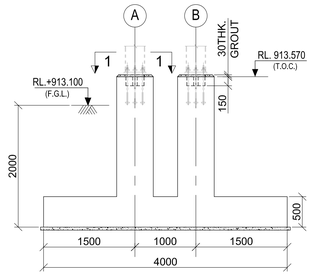




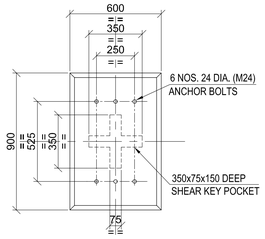
LAYOUT OF FOUNDATION
SCALE 1:100



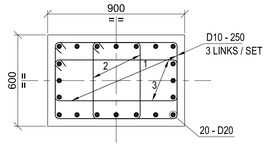
KEY PLAN
SCALE 1:1000



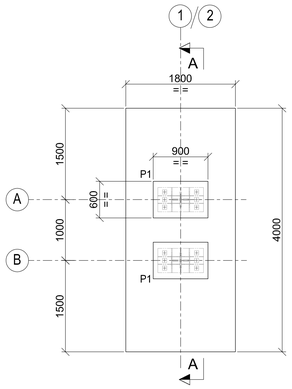
SECTION : A-A
SCALE 1:50



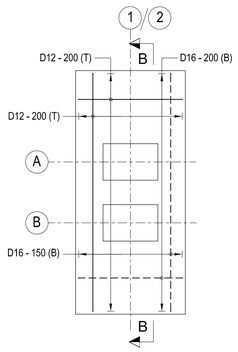
VIEW : 1-1
(DETAIL OF P1)
SCALE 1:20



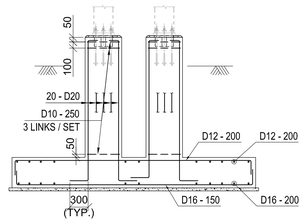
RCC. DET. OF P1
SCALE 1:20



DET. OF FOUNDATION MKD. F1
SCALE 1:50



RC. DET. OF FOUNDATION MKD. F1
SCALE 1:50



SECTION : B-B
SCALE 1:50

LEGEND :-
R.L. - REDUCED LEVEL
CL. - CENTRE LINE
I.L - INVERT LEVEL
F.G.L - FINISHED GROUND LEVEL
T.O.C. - TOP OF CONCRETE
P.C.C - PLAIN CEMENT CONCRETE
T.O.G. - TOP OF GRATING
THK. - THICK
TYP. - TYPICAL
DIA - DIAMETER
B.O.F. - BOTTOM OF FOUNDATION
DET. - DETAIL
R.C. - REINFORCED CONCRETE
MKD. - MARKED
BOTTOM REINFORCEMENT (B) - – – – – –
TOP REINFORCEMENT (T) - ————

NOTES:-
1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES UNLESS OTHERWISE STATED.
2. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION IS CONSIDERED 400KN/SQ. M.
3. GRADE OF CONCRETE SHALL BE C35 (35MPa) FOR GRADE SLAB & C15 (15MPa) FOR BLINDING CONCRETE.
4. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615 - GRADE 60 (fy = 420 MPa).
5. IF ANY PART OF FOUNDATION COMES ON BACKFILL ZONE OF ADJACENT FOUNDATION OR ANY LOOSE SOIL ARE ENCOUNTERED BELOW FOUNDATION LEVEL THE SAME SHALL BE REMOVED AND FILLED WITH PCC.
6. CLEAR COVER TO REINFORCEMENT SHALL BE FOUNDATION - 50MM., PEDESTAL - 40MM.
7. FOR STANDARD REINFORCEMENT DETAIL & GENERAL NOTES FOR CIVIL WORKS REFER DRAWING NO. MD-216-1000-EG-CV-DED-1005 & 1001.

SHEET 2 OF 2

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	21/3/2023	AS BUILT	TEC	AVG	TEC	SJM	
		C03	28/09/2021	ISSUED FOR CONSTRUCTION & REVISED AS MARKED /C03	TEC	AVG	TEC	SJM	
		C02	14/04/2021	ISSUED FOR CONSTRUCTION & REVISED AS MARKED /C02	TEC	AVG	TEC	SJM	
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL & ARCHITECTURAL - DESIGN CRITERIA	C01	27/11/2020	ISSUED FOR CONSTRUCTION	TEC	SJM	TEC	AG	
MD-216-49FE-EG-ST-GAD-1001	FUEL OIL STORAGE LPG PIPE RACK PLAN, ELEVATION & SECTION	A01	28/09/2021	ISSUED FOR APPROVAL	TEC	SJM	TEC	AG	

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OUTOTEC DRAWING NUMBER: 900707__GAU01__AG50__MTL01__49037 — DOCUMENT ID: OU100703980 — REV ON: 4

DRAWN	SYA	24/09/2021
CHECKED	SJM	27/09/2021
ENGINEER	AVG	24/09/2021
APPROVED	AG	28/09/2021
EMPLOYER		
SCALE	1:100, 1:50	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
FUEL STORAGE FACILITY-MISCELLANEOUS
PIPE SUPPORT & EQUIPMENT FOUNDATION

DRAWING NUMBER: MD - 216 - 49FE - EG - CV - GAD - 1631 — REV: D01